Nevò **MIRZAI HAMADANI**

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PROFESSIONAL SUMMARY

Robotics master's student at EPFL with deep **simulation expertise**, from modifying physics engine internals (adding fluid curing and expansion in Genesis) to training RL and **imitation learning** policies across Isaac Sim, MuJoCo, and Gymnasium. Currently planning the development of imitation learning pipelines at Embodied AI for humanoid domestic tasks, after building teleoperation infrastructure. Research interests in sim-to-real transfer, imitation learning, world models and inverse reinforcement learning, for which I derived explicit error bounds on reward recovery under approximate dynamics.

EDUCATION

EPFL | *Master in Robotics (2nd year)* Sep. 2024 – December 2026

- Minor in Data Science: Reinf. Learn.: 6/6, Deep Learn.: 5.75/6, Image analysis: 5.75/6
- Relevant courses: Robotic Manip.: 5.5/6, Aerial Robotics: 5.5/6, Legged Robots: 5.25/6

EPFL | *Bachelor in Mechanical Engineering* Sep. 2021 – July 2024

- 1 year exchange at the Chinese University of Hong Kong (CUHK)
- Relevant courses: Linear Algebra: 5.75/6, Programming: 5.75/6, Algorithms: 5.75/6

EXPERIENCE

Embodied AI | *Robotics Internship, Humanoid Learning d* Feb. 2026 – Aug. 2026

- Developing **progress-aware policies** for long-horizon bimanual manipulation, combining **SARM** reward models with **DINO-WM** world model architecture to improve reliability on smartphone charging cable insertion task.
- Building teleoperation infrastructure for **imitation learning** datasets.

EPFL | *Teaching Assistant, Student Mentor* Sep. 2022 – Dec. 2025

- Teaching Assistant for Reinforcement Learning, Algorithms, Programming, Linear Algebra, and Mechatronics; Student Mentor for freshmen.

EPFL Rocket Team | *Project Lead: Flight Simulation* Sep. 2021 – June 2024

- Led flight simulation team developing vertical landing dynamics models in Simulink.
- Explored a control method approach utilising a MPC and neural networks for better stabilization and path planning for accurate landing. Trained and tested on Isaac Sim.

RESEARCH PROJECTS

Identifiability in Inverse Reinforcement Learning — RL Course, EPFL 6.0/6.0

- Surveyed three seminal works on **IRL identifiability** (Kim et al., Cao et al., Rolland et al.) covering graph-theoretic, entropy-regularized, and multi-expert settings.
- Derived an explicit **non-asymptotic sup-norm bound** on the reward estimation error as a function of the spectral deviation between true and empirical transition matrices, extending findings from Rolland et al.
- Provided **sample complexity bounds** for reward recovery.

3D Printing Drone Simulation — EMPA Lab Zurich, Prof. Kovac — Paper Pending

- Built a simulation framework for an aerial additive manufacturing drone in the **Genesis** physics engine.
- Modified the engine's core **MPM (Material Point Method)** solver to model **viscoelastic curing and expanding fluid dynamics** — beyond standard rigid-body simulation.
- Validated simulated material deposition behavior against physical experiments.

Aerial Swarm Control via Hand Gestures — LIS Lab, EPFL, Prof. Floreano

- Designed an intuitive hand-gesture interface for controlling a drone swarm via a VR headset, with a modified **Olfati-Saber flocking algorithm** for swarm coordination.
- Built a simulated swarm in Unity and transferred control to a **real Crazyflie swarm via ROS2** (Sim-2-Real).

SELECTED PROJECTS

Imitation Learning with Robotic Arm — ACT, Vision-Based Control

- Assembled a leader-follower robotic arm system for teleoperated data collection.
- Implemented an **ACT (Action Chunking with Transformers)** policy for multi-step manipulation.
- Designed a vision-based control pipeline with **ResNet-18** encoders processing dual camera streams.
- Achieved ~**85% success rate** on pick-and-place tasks from 50 demonstrations.

Reinforcement Learning for Quadruped Locomotion — SAC, Isaac Gym

- Trained a **SAC** reinforcement learning policy in **Isaac Gym** for quadruped locomotion.
- Achieved multi-gait control (trot, walk, pace, bound) with velocity tracking and **stair climbing** up to 5 cm step height.
- Designed **reward functions** for forward progress, smoothness, and drift penalty; used **adaptive terrain curriculum** for progressive difficulty.

Autonomous Drone Racing — Sim2Real, Path Planning

- Programmed autonomous trajectory racing in simulation, then deployed on a real Crazyflie via **sim-to-real transfer**.
- Integrated path planning, state estimation, and computer vision.

Data Creation Box — Object Recognition, System Integration - 3rd Place Startup Competition

- Built an automated physical environment generating diverse training datasets; fine-tuned object recognition models deployed on a robotic arm.

TECHNICAL SKILLS

Programming Python, C++, C, MATLAB — **Technical** ROS2, Docker, CUDA, Git
Simulation IsaacSim, Mujoco, Gazebo, Simulink, Gymnasium, Unity, Genesis, Webots
Machine Learning SAC, PPO, ACT, Behavioral Cloning, IRL, PyTorch, IsaacLab
Computer Vision SLAM (Visual/LiDAR), Image Analysis, OpenCV
Laboratory Soldering, electrical circuits, CAD, 3D Printing

AWARDS - INTERESTS - LANGUAGES

Guinness World Record (2017): Most drones built and flown simultaneously (U18)
Talent Kick (2026): Selective cross-university high-tech startup incubator.
English C1, French C1, Italian C2, German B1 — Judo (competed internationally)